



TECHNICAL DATA SHEET

— Parasmani MGPS Copper Tubes

Seamless Copper Tubes for Medical Gas Pipeline Systems and Vacuum

I. SCOPE & APPLICABLE STANDARDS

This technical data sheet covers Parasmani MGPS phosphorus-deoxidized (DHP) seamless round copper tubes manufactured specifically for Medical Gas Pipeline Systems (MGPS) and medical vacuum services, intended for the safe and reliable distribution of medical gases in accordance with the applicable standards listed in this document.

Standard	Application
BS EN 13348	Medical gases and vacuum — primary medical gas standard for the tube
ASTM B819	Medical gas tube — relevant requirements met

Where a project specification calls for a standard other than the above, requirements shall be confirmed in writing prior to manufacture.

2. MATERIAL & CHEMICAL COMPOSITION

Parasmani MGPS copper tubes are manufactured from high-purity C12200 DHP (phosphorus-deoxidized) copper, designated CW024A, with a minimum copper content of 99.90% (including silver). The controlled chemical composition ensures excellent thermal conductivity, corrosion resistance, formability, weldability, and consistent mechanical properties throughout the full cross-section, making the material suitable for Medical Gas Pipeline Systems (MGPS) and medical vacuum installations.

Characteristic	Value
Material Grade	C12200 (DHP) / CW024A
Copper Content	Cu 99.90% minimum (including silver)
Phosphorus (deoxidizer)	0.015–0.040%
Total Impurities	Maximum 0.060% (excluding phosphorus and silver)
Copper Density	8.94 g/cm ³
Copper Melting Point	1084.6 °C



3. MECHANICAL & METALLURGICAL PROPERTIES

Property	Requirement
Temper (standard supply)	Half Hard (R250)
Temper (on request)	Hard (R290)
Tensile Strength	250 MPa minimum (R250)
Elongation	30% minimum
Vickers hardness, HV 5 (indicative)*	75 to 100
Drift expanding test	No cracking on the surface
Flattening test	No cracking on the surface
Eddy current test	No detrimental flaw
Residual carbon, internal surface**	0.20 mg/dm ² maximum
Surface condition	Clean, smooth and free from scale, loose particles, oil, grease and moisture

* Hardness is stated for guidance only and is not a requirement of the standard.

** Determined using a carbon determinator in accordance with the reference method specified in EN 723.

Why Half Hard: R250 is the temper recommended under BS EN 13348 for medical gas pipelines and is therefore Parasmani's standard supply. It resists long-term stress cracking, absorbs vibration and hammering rather than transmitting it to the joints, and accommodates thermal and seismic movement — where hard drawn R290 is less tolerant.

4. MANUFACTURING ROUTE & INTERNAL CLEANLINESS

Tubes are produced by extrusion and cold drawing. No welded, brazed or seamed construction is used at any stage. The process sequence is: melting and casting → extrusion → drawing → cutting → straightening → annealing → visual inspection → eddy current and allied testing → packing.

For medical gas service, cleanliness of the bore is the governing requirement:

- Every tube is internally degreased after the final draw pass, thoroughly dried and securely end-capped before leaving the factory.
- Internal cleanliness is verified to the BS EN 13348 limit of 0.20 mg/dm² total residual carbon.
- Tubes are free from oil, grease, drawing lubricant, moisture, scale and metallic particles on both internal and external surfaces, and free from odour in the as-supplied condition.
- End caps remain in place through transport and storage and are removed only at the point of installation.



5. NINE-STEP QUALITY ASSURANCE

Each lot is subjected to the following in-house test regime before despatch:

01	Chemical composition analysis	06	Drift expanding test
02	Hydrostatic pressure test	07	Flattening test
03	Eddy current test	08	Hydrogen embrittlement test
04	Freedom from defects	09	Internal cleanliness test
05	Pneumatic pressure test		

6. DIMENSIONS, TOLERANCES & PACKING

PARASMANI MGPS STRAIGHT COPPER TUBES STANDARD PIPE LENGTH: 3 MTR						
O.D. (mm)	W.T. (mm)	O.D. Tolerance	W.T. Tolerance	Weight Per Mtr. (kg)	Pipes Per Bundle	Weight Per Bundle
10	0.9	± 0.09	± 0.09	0.231	50	34.65
12	0.7	± 0.09	± 0.07	0.223	50	33.45
12	1.0	± 0.09	± 0.10	0.309	40	37.08
15	0.7	± 0.09	± 0.07	0.282	40	33.84
15	0.9	± 0.09	± 0.09	0.357	40	42.84
15	1.0	± 0.09	± 0.10	0.394	35	41.37
22	0.9	± 0.10	± 0.09	0.534	25	40.05
22	1.0	± 0.10	± 0.10	0.590	20	35.40
28	0.9	± 0.10	± 0.09	0.686	20	41.16
28	1.0	± 0.10	± 0.10	0.759	15	34.16
35	1.2	± 0.11	± 0.12	1.140	10	34.20
42	1.2	± 0.11	± 0.12	1.376	10	41.28
54	1.2	± 0.11	± 0.12	1.780	8	42.72
76.1	1.5	± 0.15	± 0.225	3.143	4	37.72
108	1.5	± 0.30	± 0.225	4.487	3	40.38

Custom wall thicknesses, cut-to-length and additional sizes are available on request – including OD 70 mm and 80 mm, and 108 × 2.5 mm. Maximum outside diameter offered is 108 mm.

Standard pipe length	3 mtr, (3000 mm) straight lengths
Length tolerance	+ 10 mm / – 0 mm, unless otherwise agreed at the time of order
Packing	Segregated by size, wall thickness and temper; both ends capped; bundled and crated for despatch by common carrier



7. TUBE MARKING & TRACEABILITY

Every straight length carries a permanent surface marking applied by laser print / electro-etching, together with a Parasmani security hologram. The marking sequence is:

PARASMANI MGPS · EN 13348 · HH / R250 · Cu DHP · 15 mm × 1.0 mm · PT120875

Read in order: brand name · standard · temper · grade · outside diameter × wall thickness · batch number. The batch number is traceable to the mill test certificate issued with the consignment.

8. JOINTING & INSTALLATION GUIDANCE

Braze with CP 104 copper-silver-phosphorus filler to BS EN ISO 17672 — no flux is required on copper-to-copper joints. Indicative minimum brazing temperature is 710 °C. Penetration must reach three times the wall thickness or 3 mm, whichever is greater.

Nitrogen shielding. Oxygen-free nitrogen must flow through the pipework throughout brazing to prevent oxide scale forming on the bore. Flush first to clear air, maintain flow while brazing, and purge afterwards to remove shield gas and particulate. This preserves the internal cleanliness achieved during manufacture — the condition on which oxygen service depends.

Ensure the torch reaches brazing temperature across the full joint; oxy-acetylene or oxygen-fed LPG is recommended, as air-fed LPG may not reach temperature on larger sizes. Avoid prolonged heating. In confined spaces, ventilate to prevent nitrogen build-up and monitor ambient oxygen content.

9. CERTIFICATION & DOCUMENTATION

A Manufacturer's Test Certificate showing the results of the tests required by the specification accompanies every despatch and is traceable to the batch number marked on the tube.

10. PRODUCT ASSURANCE

All copper tubes supplied against order shall conform to the specification set out above and to the specific requirements of the purchase order. Any deviation required by a project specification shall be mutually agreed in writing prior to manufacture. This sheet is issued for technical reference; commercial terms, pricing and delivery are covered separately in our quotation.

A project-specific technical submittal on our letterhead, together with mill test certificates, is issued on request.